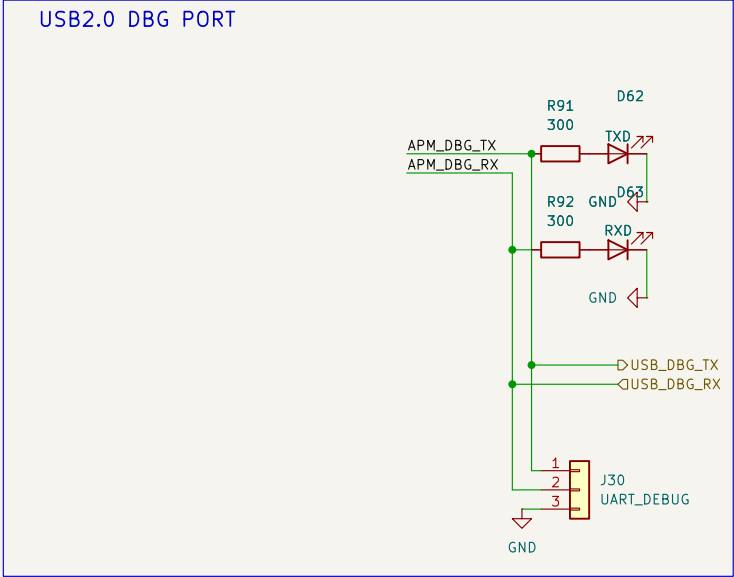


UART -> USB Debug Port

UART debug messages available via a USB port and a 3 pin jumper for TX, RX, and GND UART connections



Sheet: /USB_debug_APM/
File: USB_debug_APM.kicad_sch

Title:

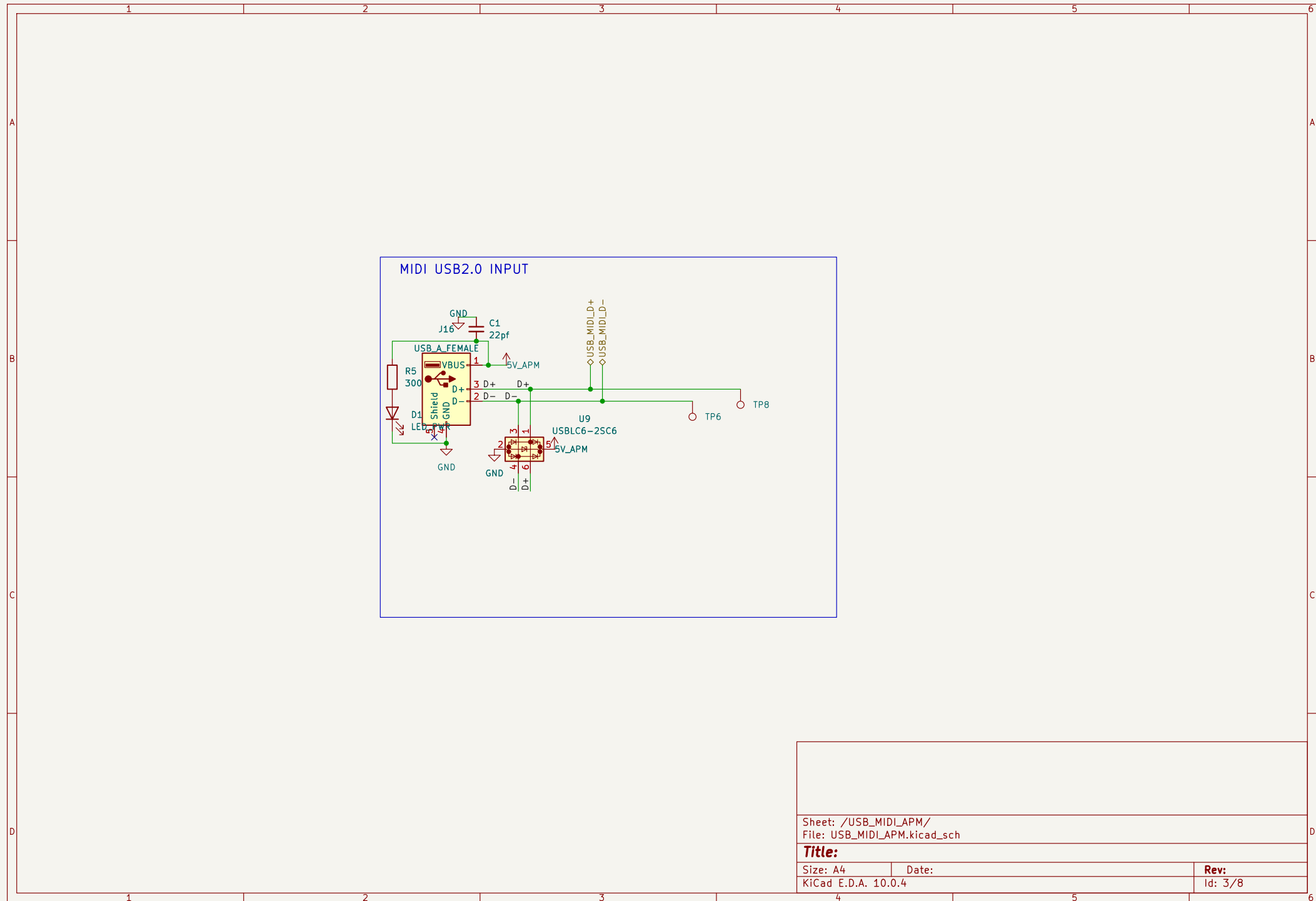
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Date:

KiCad E.D.A. 10.0.4

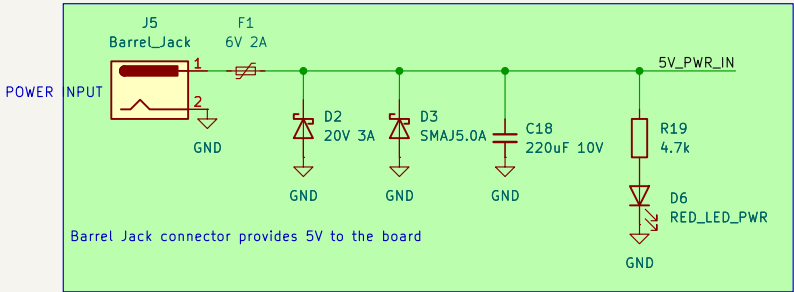
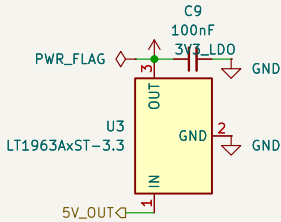
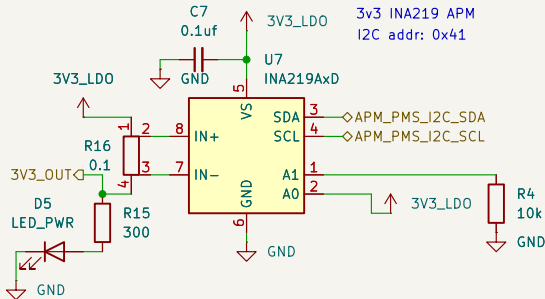
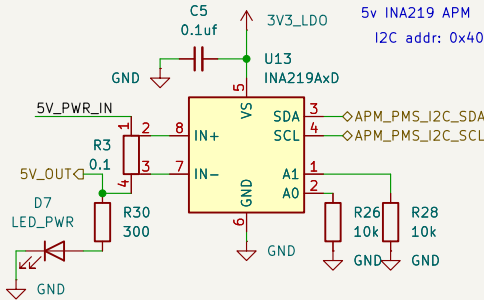
Rev:

Id: 2/8



Power Monitor and Supply

PMS has an INA219 current sensor for each board/node in the PCB. Each takes in the 5V from it's parent board and supplies 5V to said board's circuit (its peripherals) thus enabling monitoring



Barrel Jack connector provides 5V to the board

Sheet: /PMS_APM/
File: PMS_APM.kicad_sch

Title:

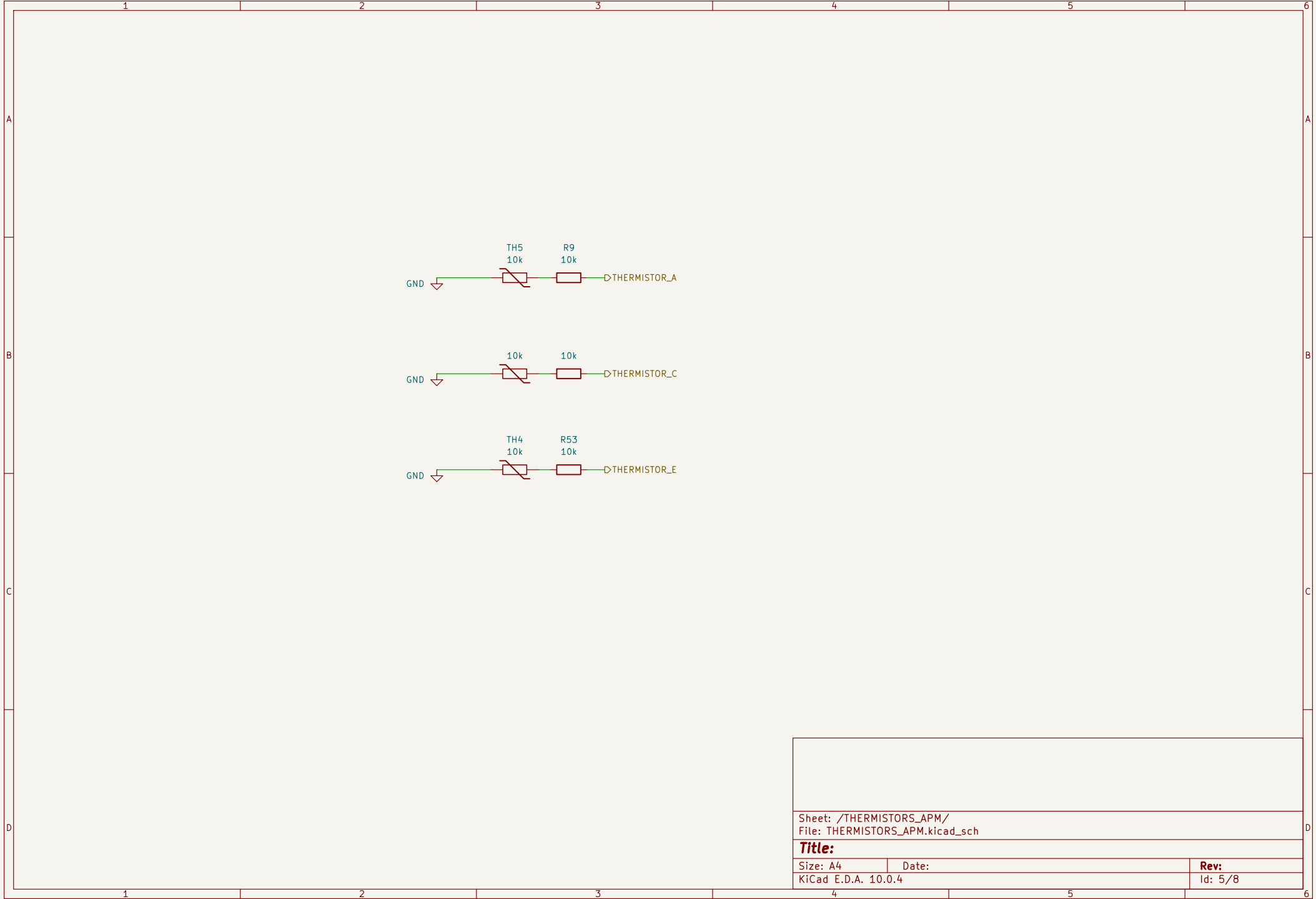
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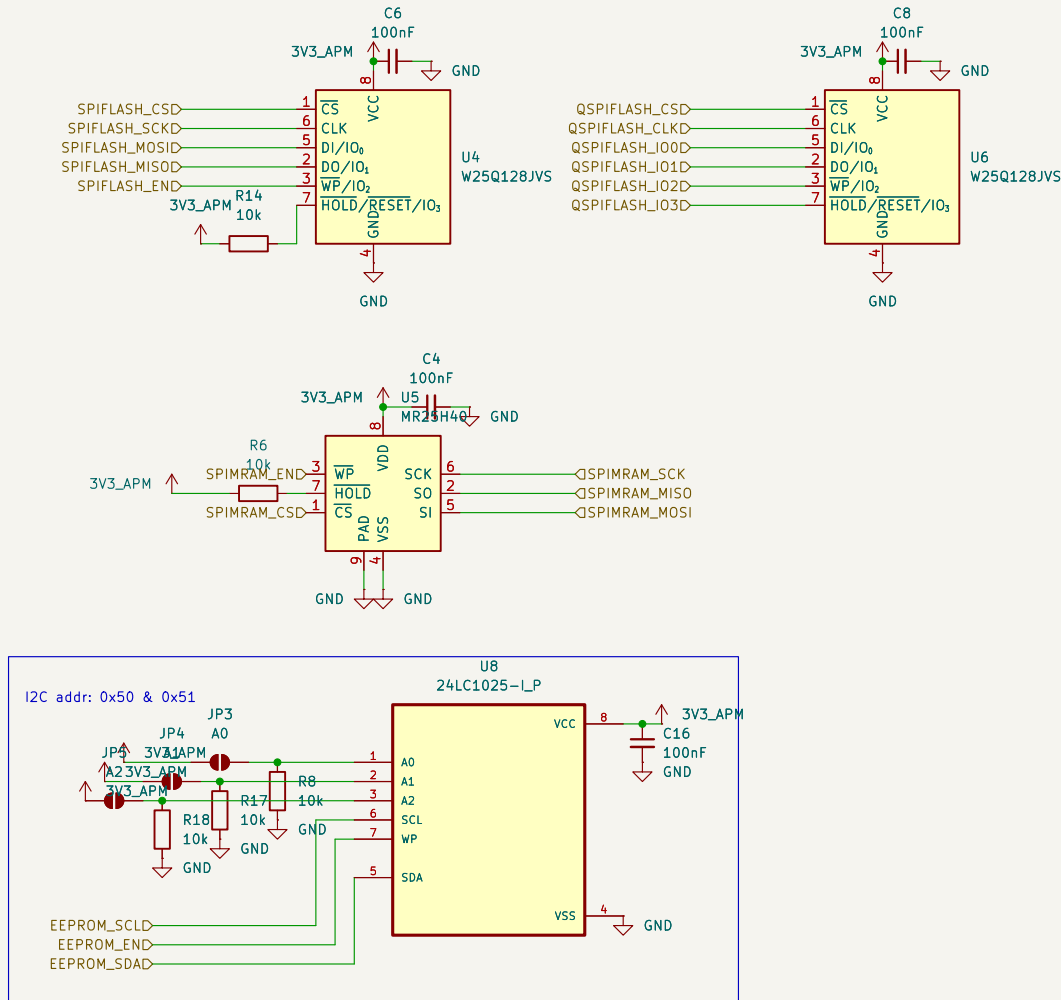
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Rev:

Id: 4/8





Sheet: /STORAGE_APM/
 File: STORAGE_APM.kicad_sch

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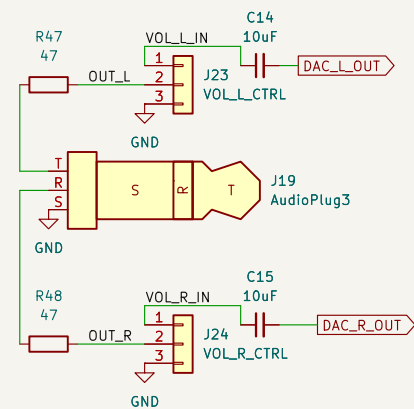
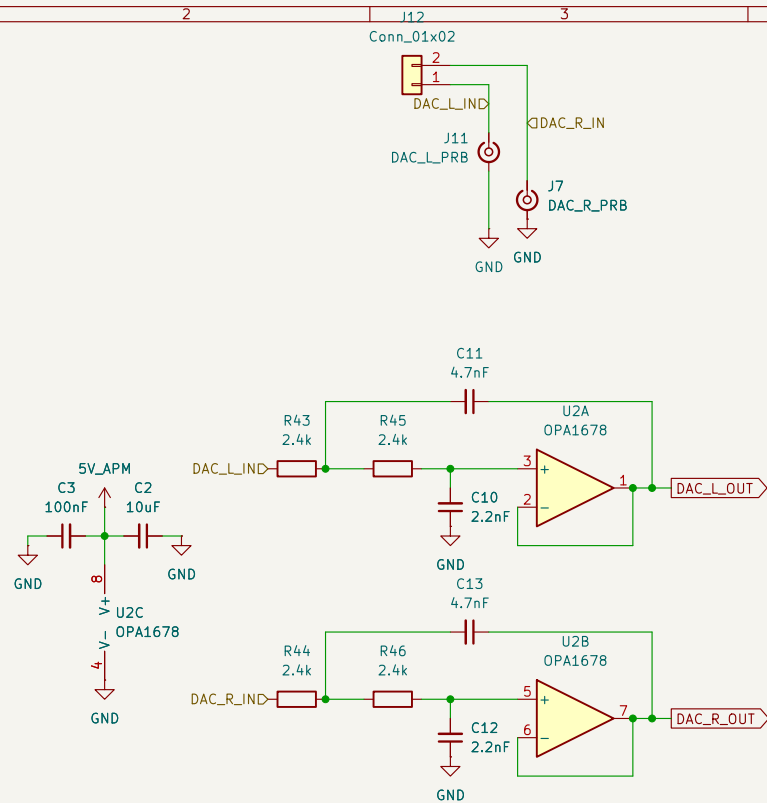
Size: A4

Date:

Rev:

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Id: 6/8



Sheet: /AUDIO_APM/
File: AUDIO_APM.kicad_sch

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Size: A4

Date:

KiCad E.D.A. 10.0.4

Rev:

Id: 7/8

ACM FPGA Interface

Dual FPC STM32<->Tang Nano 20K FPGA connection configuration. The STM32 side has two connectors where only one can be active at a time

- Configuration 1: FPGA_FFC with SPI, QSPI, UART connections.
- Configuration 2: FPGA_FMC with:
 - 16 bit muxed data
 - and async FMCconfiguration resulting in a 64MB shared address space between the STM32 & FPGA

